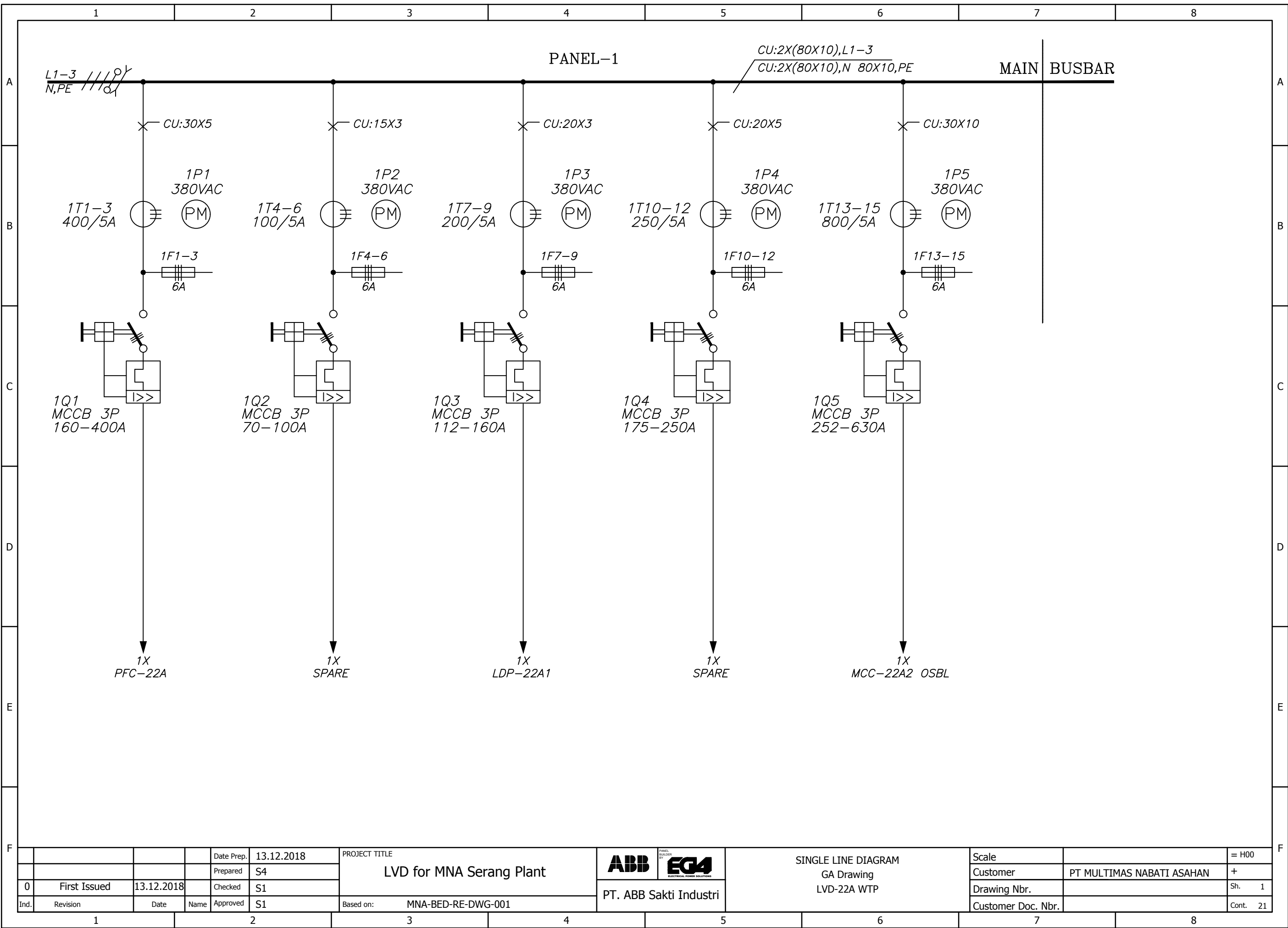


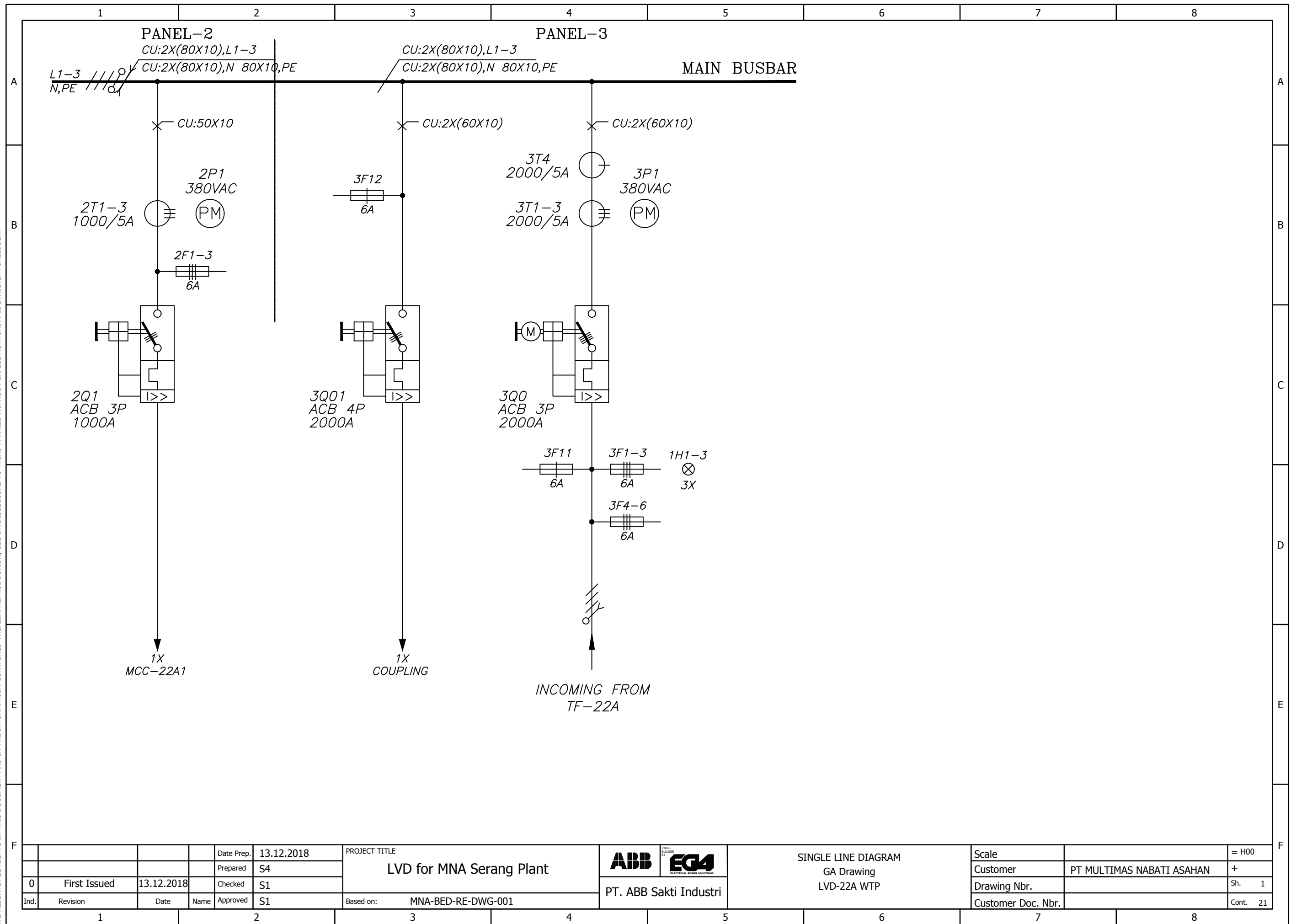
			Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	 	FRONT VIEW GA Drawing LVD-22A WTP	Scale		= H00
			Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
First Issued	13.12.2018		Checked	S1				Drawing Nbr.		Sh. 1
Revision	Date	Name	Approved	S1	Based on: MNA-BED-RE-DWG-001	PT. ABB Sakti Industri		Customer Doc. Nbr.		Cont. 21

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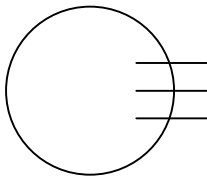
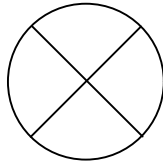
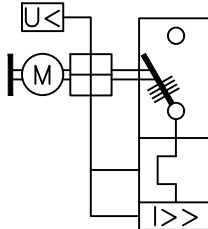
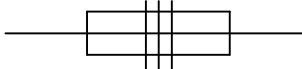
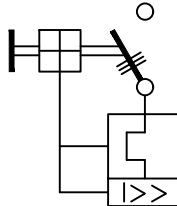
		Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant		  ELECTRICAL POWER SOLUTIONS	SINGLE LINE DIAGRAM GA Drawing LVD-22A WTP		Scale		= H00
		Prepared	S4						Customer	PT MULTIMAS NABATI ASAHAN	+
0		First Issued	13.12.2018	Checked	S1				Drawing Nbr.		Sh. 1
Ind.	Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001	PT. ABB Sakti Industri		Customer Doc. Nbr.	Cont. 21

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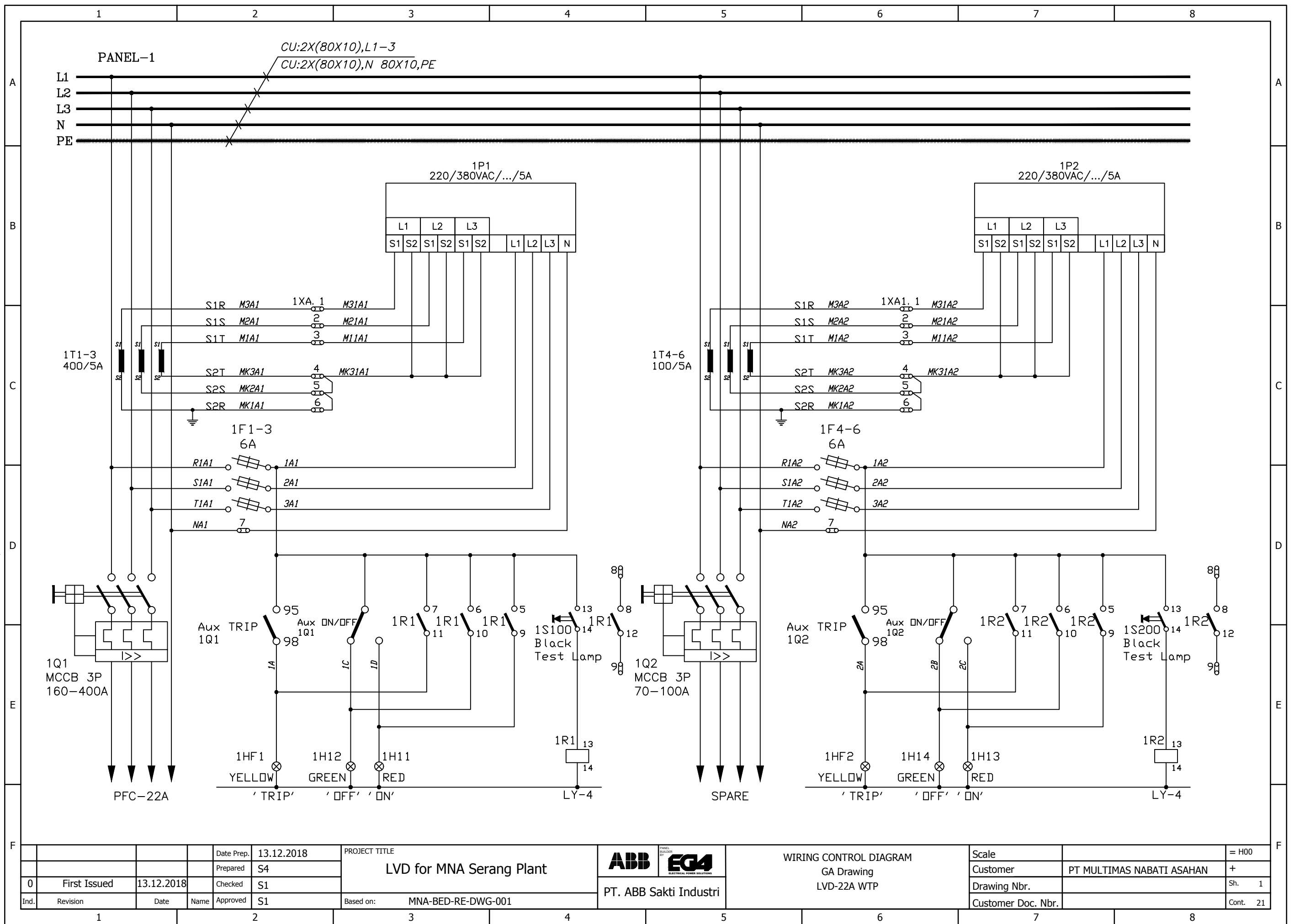


		Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	ABB PANEL SELECTOR BY EG4 PT. ABB Sakti Industri	SINGLE LINE DIAGRAM GA Drawing LVD-22A WTP	Scale	= H00	
		Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
0		First Issued	13.12.2018				Drawing Nbr.		Sh. 1
Ind.	Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001	Customer Doc. Nbr.	Cont. 21

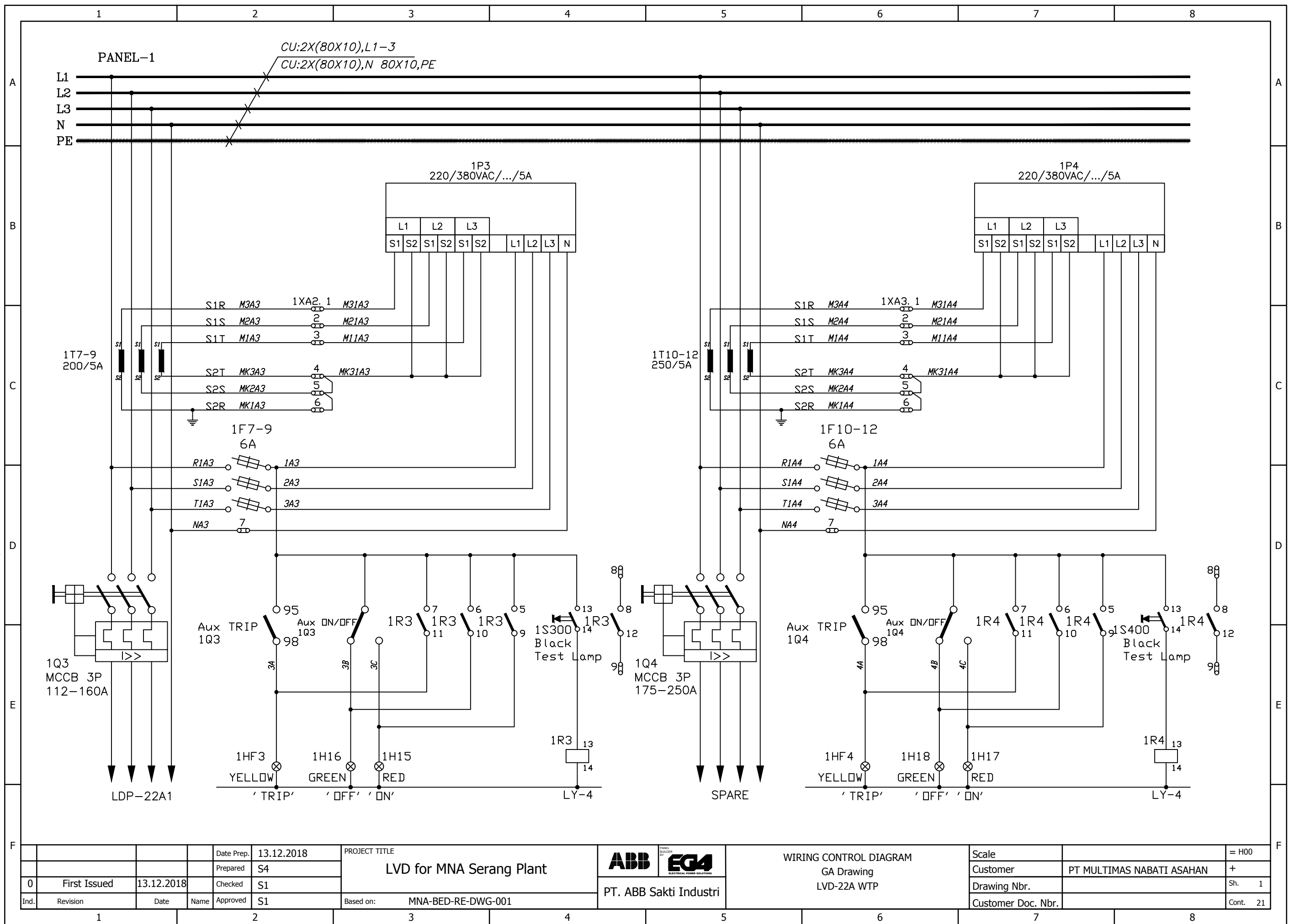
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1		2		3		4		5		6		7		8			
A																A	
		 CT  PILOT LAMP															
B																B	
		 ACB  FUSE															
C																C	
		 MCCB															
D																D	
		 DIGITAL POWER METER															
E																E	
F																F	
				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant				ABB PANEL BUILDER BY EG4 ELECTRICAL POWER SOLUTIONS PT. ABB Sakti Industri	LEGEND GA Drawing LVD-22A WTP		Scale		= H00		
				Prepared	S4								Customer	PT MULTIMAS NABATI ASAHAN	+		
0	First Issued	13.12.2018		Checked	S1								Drawing Nbr.		Sh.	1	
Ind.	Revision	Date	Name	Approved	S1	Based on: MNA-BED-RE-DWG-001						Customer Doc. Nbr.		Cont.	21		
1		2		3		4		5		6		7		8			

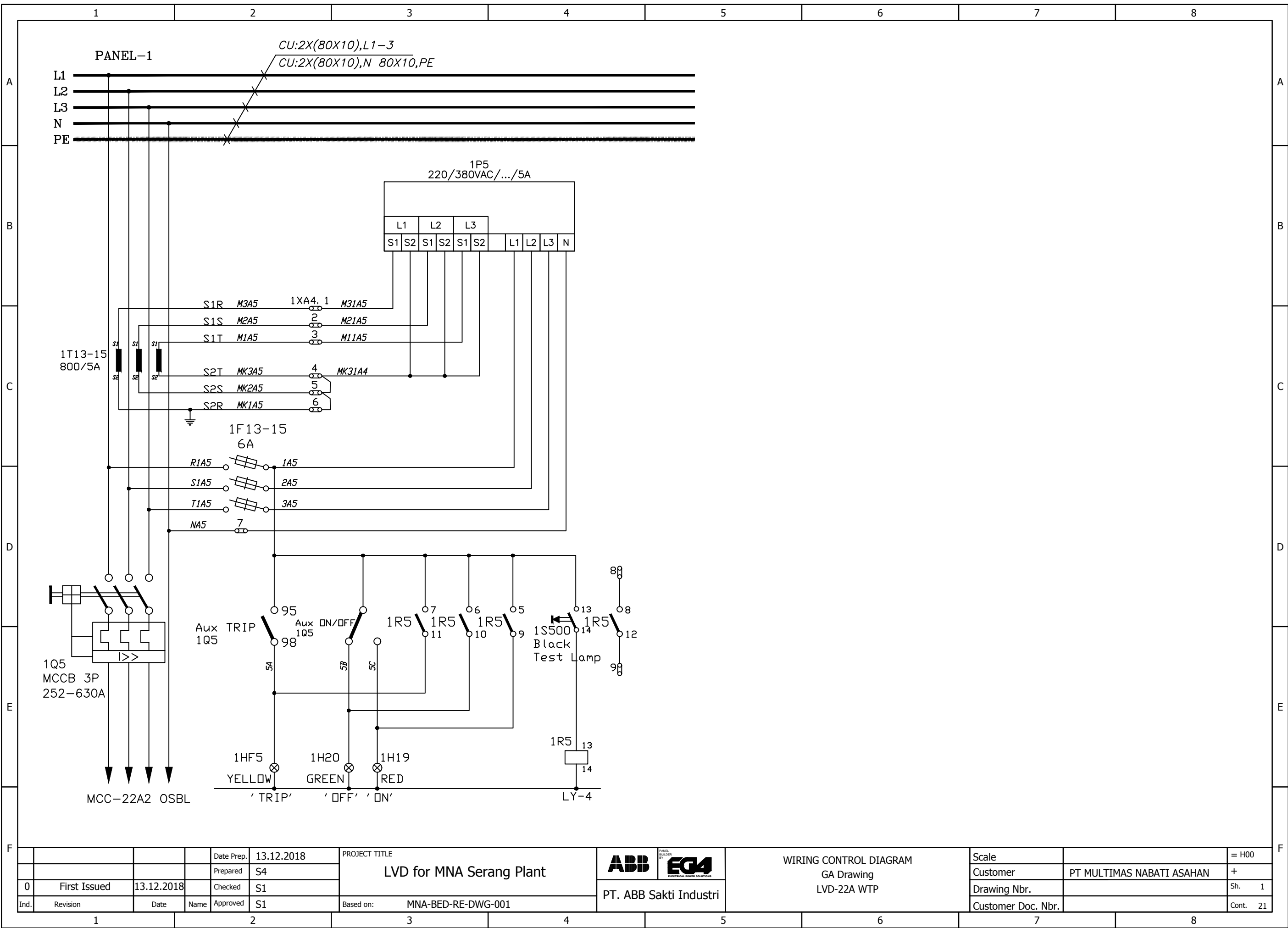
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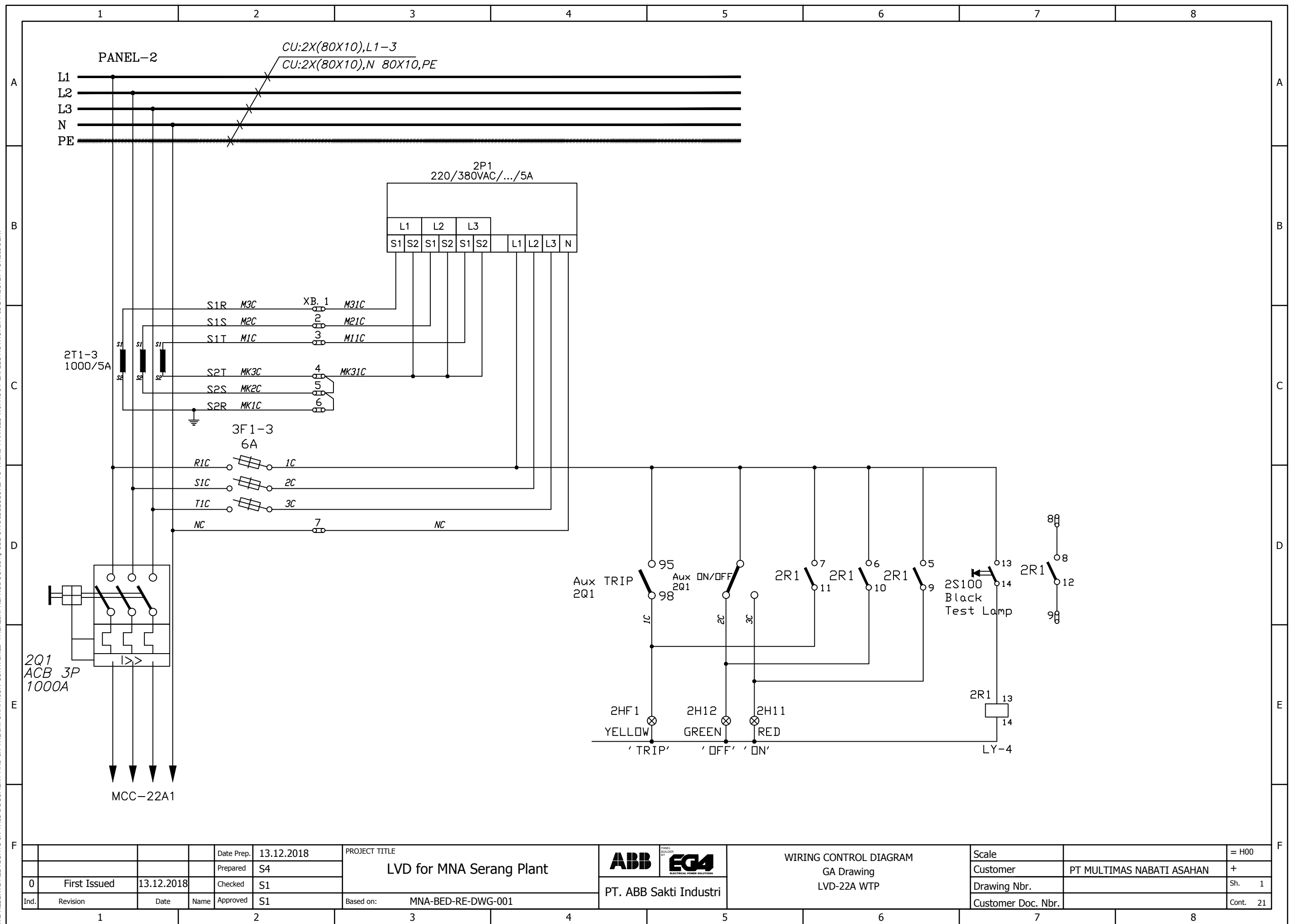
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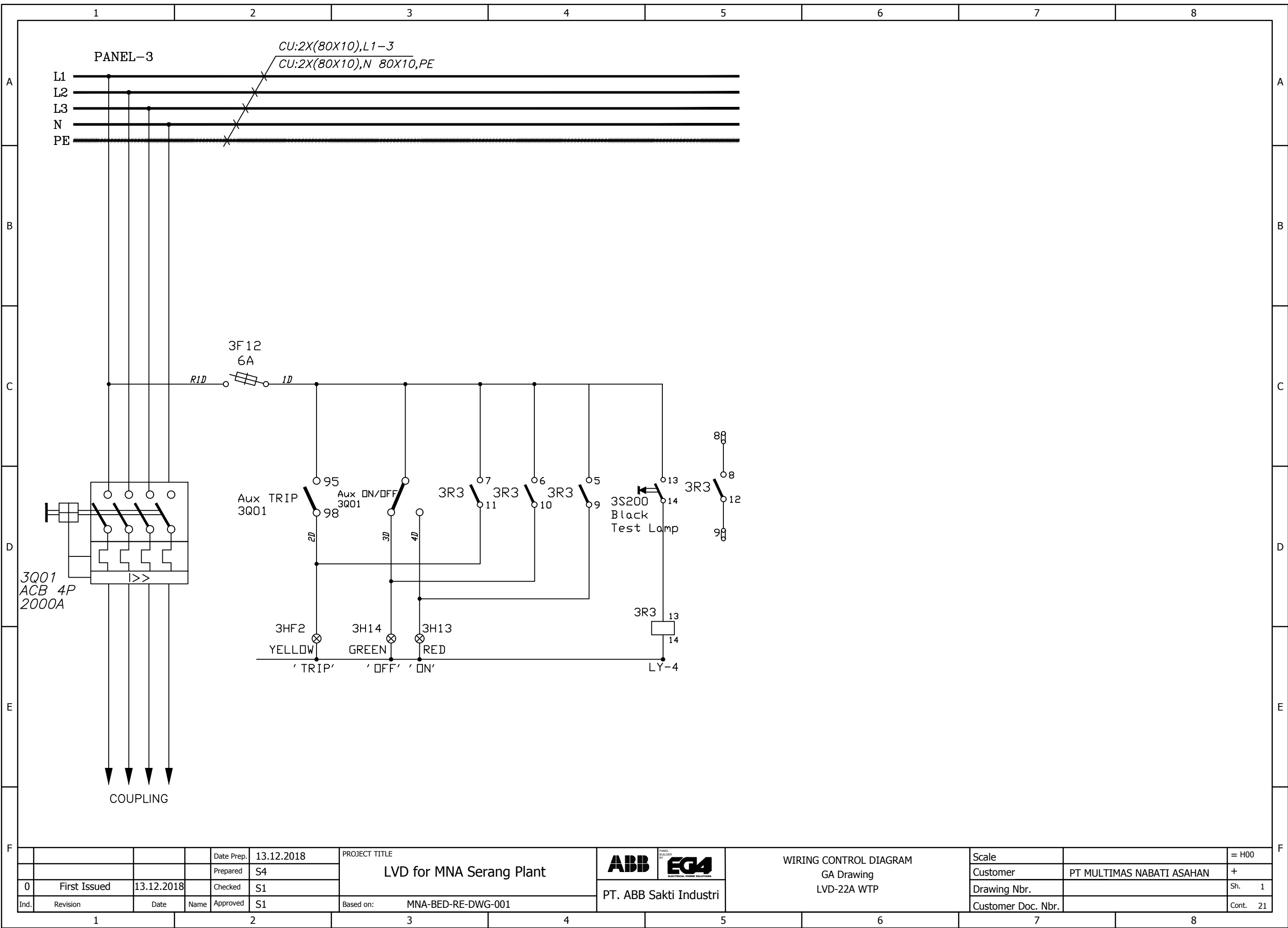
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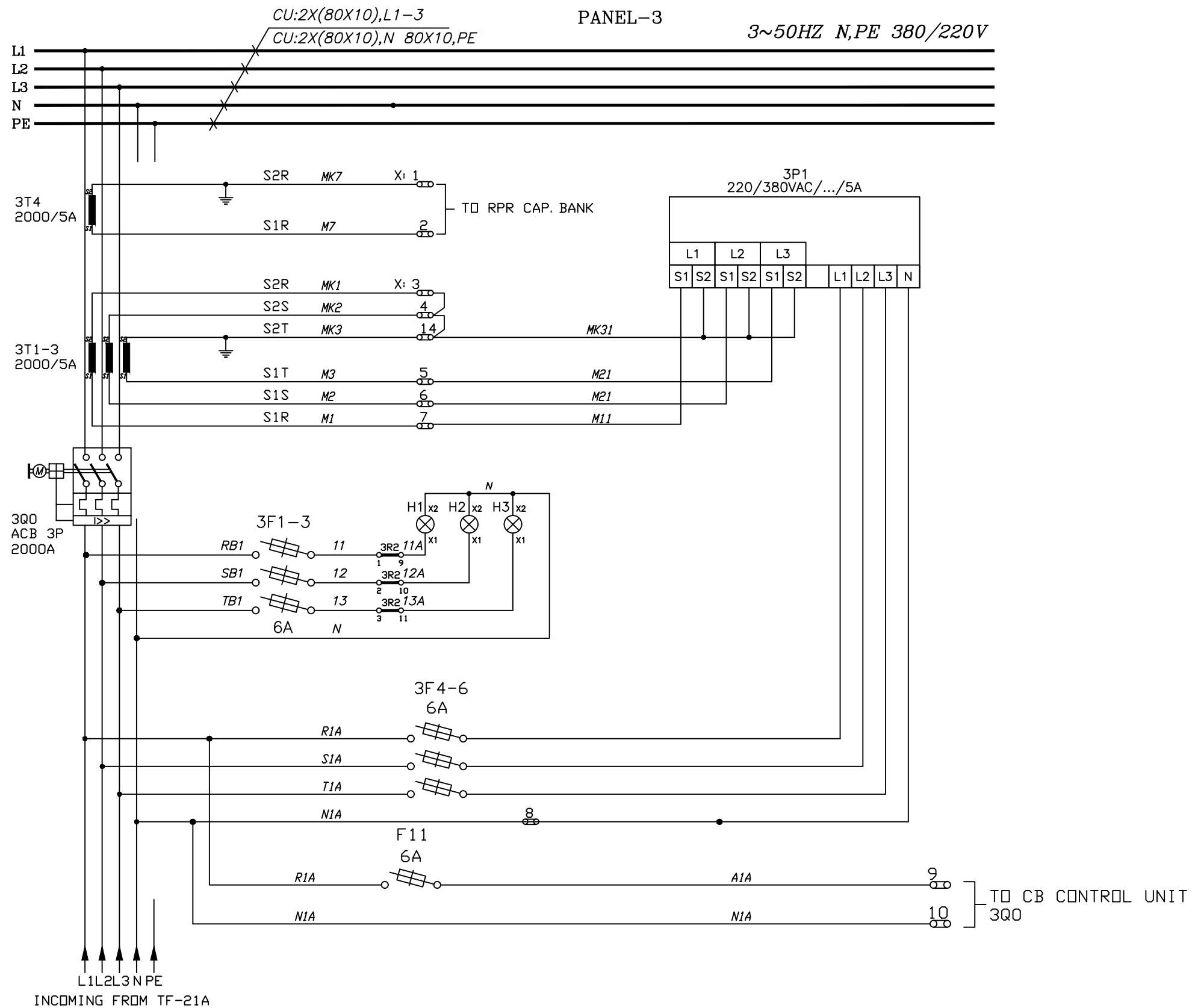
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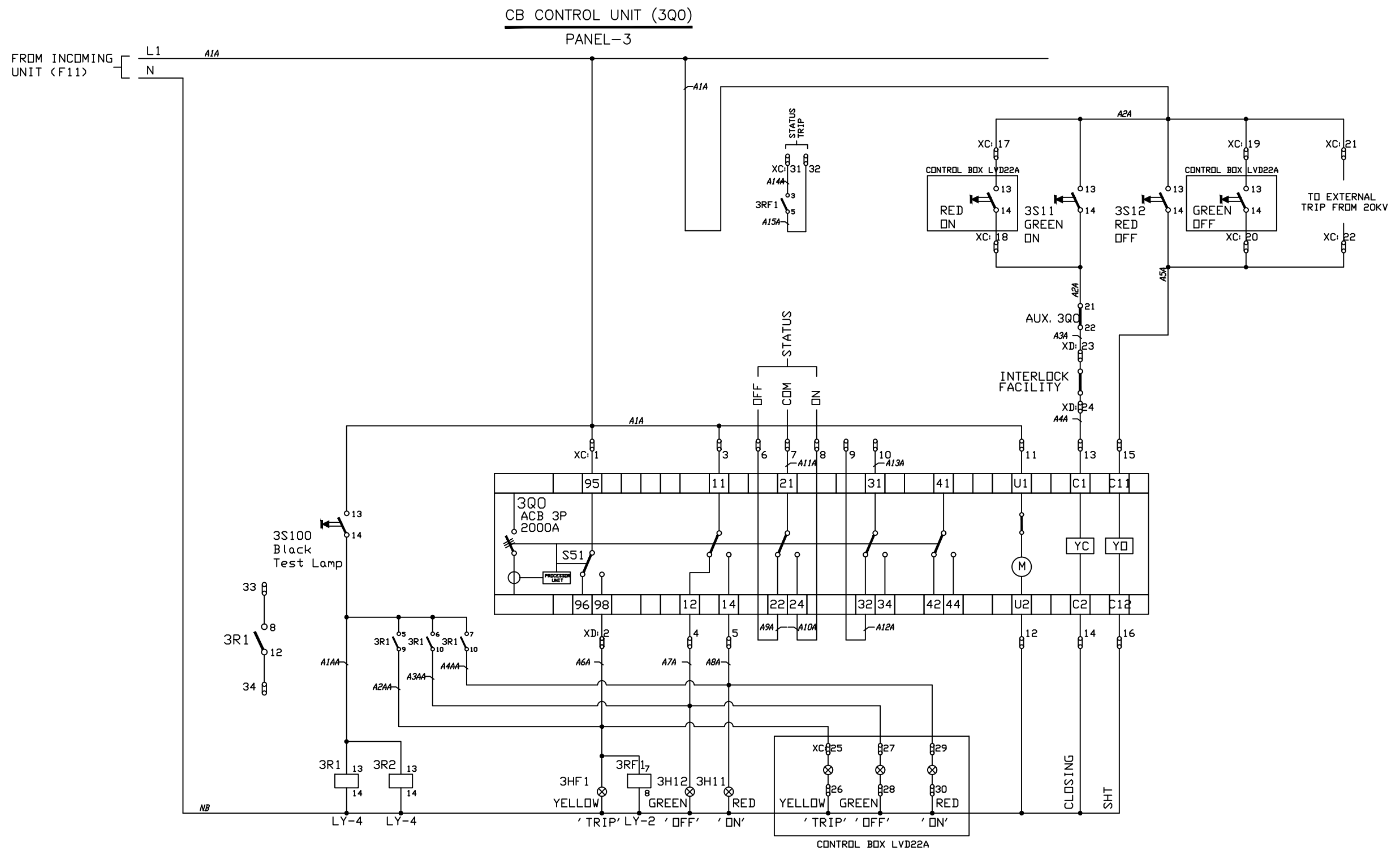


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		Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	  ELECTRICAL POWER SOLUTIONS	WIRING CONTROL DIAGRAM GA Drawing LVD-22A WTP	Scale		= H00
		Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
0		First Issued	13.12.2018				Drawing Nbr.		Sh. 1
Ind.	Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001	Customer Doc. Nbr.	Cont. 21

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A	N O		DESCRIPTION		TECHNICAL DATA		QTY		DESIGNATION						A			
			INCOMING															
B	1	Fuse Carrier 6A		AXLE/Setara		4		3F11,1F1-3						B				
	2	Pilot Lamp (LED) 220Vac/50Hz, Red		AXLE/Setara		2		3H1,11										
	3	Pilot Lamp (LED) 220Vac/50Hz, Yellow		AXLE/Setara		2		3H2,HF1										
	4	Pilot Lamp (LED) 220Vac/50Hz, Green		AXLE/Setara		2		3H3,H12										
	5	DIGITAL POWER METER (*)		ABB, M2M ETHERNET		1		3P1										
	6	CT 2000/5A CL.0.5 20VA		Howig, HG 82-2000		4		3T1-4										
C	7	ACB 3P 2000A, 66KA, DO, Ekip Touch LSIG (*)		ABB, E2.2N20 D/O		1		3Q0						C				
	8	+ Shunt Trip Release (Y0) (*)		ABB, Y0		1		-										
	9	+ Motor (M) + Closing Release (YC) For E2.2-E6.2 (*)		ABB, M+YC		1		-										
	10	+ Padlocks For E2.2-E6.2 (*)		ABB,		1		-										
	11	+ Ekip Com Protocol IEC 61850 (*)		ABB,		1		-										
	12	Push Button On, 1NO, Green		Ega		1		S11										
D	13	Push Button Off, 1NO, Red		Ega		1		S12						D				
	14	Control Relay 220Vac		Omron, LY-2		1		RF1										
	15	Control Relay 220Vac		Omron, LY-4		2		R1-2										
	16	Push Button Test Lamp , 1NO, Black		Ega		1		S100										
	17																	
	18	COUPLER																
E	19	Fuse Carrier 6A		AXLE/Setara		2		2F8						E				
	20	Pilot Lamp (LED) 220Vac/50Hz, Red		AXLE/Setara		1		3H14										
	21	Pilot Lamp (LED) 220Vac/50Hz, Yellow		AXLE/Setara		1		3HF2										
	22	Pilot Lamp (LED) 220Vac/50Hz, Green		AXLE/Setara		1		3H13										
	23	ACB 4P 2000A, 66KA, FT, Ekip Dip LI (*)		ABB, E2.2N20 F/T		1		3Q01										
	24	Push Button On/Test Lamp , 1NO, Black		Ega		1		S200										
	25	Control Relay 220Vac		Omron, LY-4		1		R3										
	26																	
F					Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant		  ELECTRICAL POWER SOLUTIONS	COMPONENT LIST GA Drawing LVD-22A WTP		Scale		= H00		F		
					Prepared	S4						Customer		PT MULTIMAS NABATI ASAHAN			+	
	0	First Issued	13.12.2018		Checked	S1						Drawing Nbr.					Sh. 1	
	Ind.	Revision	Date	Name	Approved	S1	Based on: MNA-BED-RE-DWG-001		PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 21				
	1		2		3		4		5		6		7		8			

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1		2		3		4		5		6		7		8	
A	N O		DESCRIPTION		TECHNICAL DATA		QTY		DESIGNATION						A
			OUTGOING												
B	1	Fuse Carrier 6A		AXLE		18		1F1-15 ; 2F1-3						B	
	2	Pilot Lamp (LED) 220Vac/50Hz, Red		AXLE		6		1H11,13,15,17,19 ; 2F11							
	3	Pilot Lamp (LED) 220Vac/50Hz, Yellow		AXLE		6		1HF1-5 ; 2HF1							
	4	Pilot Lamp (LED) 220Vac/50Hz, Green		AXLE		6		1H12,14,16,18,20 ; 2F12							
	5	DIGITAL POWER METER (*)		ABB, M2M ETHERNET		6		1P1-5 ; 2P1							
	6	CT 1000/5A CL.0.5 10VA		Howig, HG 82-1000		3		2T1-3							
C	7	CT 800/5A CL.1 10VA		Howig, HG 50I-800		3		1T13-15						C	
	8	CT 400/5A CL.1 5VA		Howig, HG 40I-400		3		1T1-3							
	9	CT 250/5A CL.1 5VA		Howig, HG 30I-250		3		1T10-12							
	10	CT 200/5A CL.1 5VA		Howig, HG 30I-200		3		1T7-9							
	11	CT 100/5A CL.1 2.5VA		Howig, HG 30I-100		3		1T4-6							
	12	ACB 3P 1000A, 50KA, FT, Ekip Dip LI (*)		ABB, E1.2C10 F/T		1		2Q1							
D	13	+ Padlocks For E1.2 (*)		ABB,		1		-						D	
	14	MCCB 3P 252-630A, 50KA (*)		ABB, T5S PR221DS-LS/I-630		1		1Q5							
	15	+ Auxiliary Contact For T5-T6 (*)		ABB, Aux-C 3Q 1SY		1		-							
	16	+ Extended Rotary Handles For T5 (*)		ABB, RHE		1		-							
	17	MCCB 3P 160-400A, 50KA (*)		ABB, T5S PR221DS-LS/I-400		1		1Q1							
	18	+ Auxiliary Contact For T5-T6 (*)		ABB, Aux-C 3Q 1SY		1		-							
E	19	+ Extended Rotary Handles For T5 (*)		ABB, RHE		1		-						E	
	20	MCCB 3P 175-250A, 50KA (*)		ABB, XT3S TMD 250		1		1Q4							
	21	+ Auxiliary Contact For XT2/XT4 (*)		ABB, Aux-C 2Q 1SY		1		-							
	22	+ Extended Rotary Handles For XT2/XT4 (*)		ABB, RHE		1		-							
	23	MCCB 3P 112-160A, 50KA (*)		ABB, XT2S TMA-160		1		1Q3							
	24	+ Auxiliary Contact For XT2/XT4 (*)		ABB, Aux-C 2Q 1SY		1		-							
F	25	+ Extended Rotary Handles For XT2/XT4 (*)		ABB, RHE		1		-						F	
	26														
1		2		3		4		5		6		7		8	

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A <															